A scalable metagenomic approach for joint quantification of gut microbiome and dietary intake

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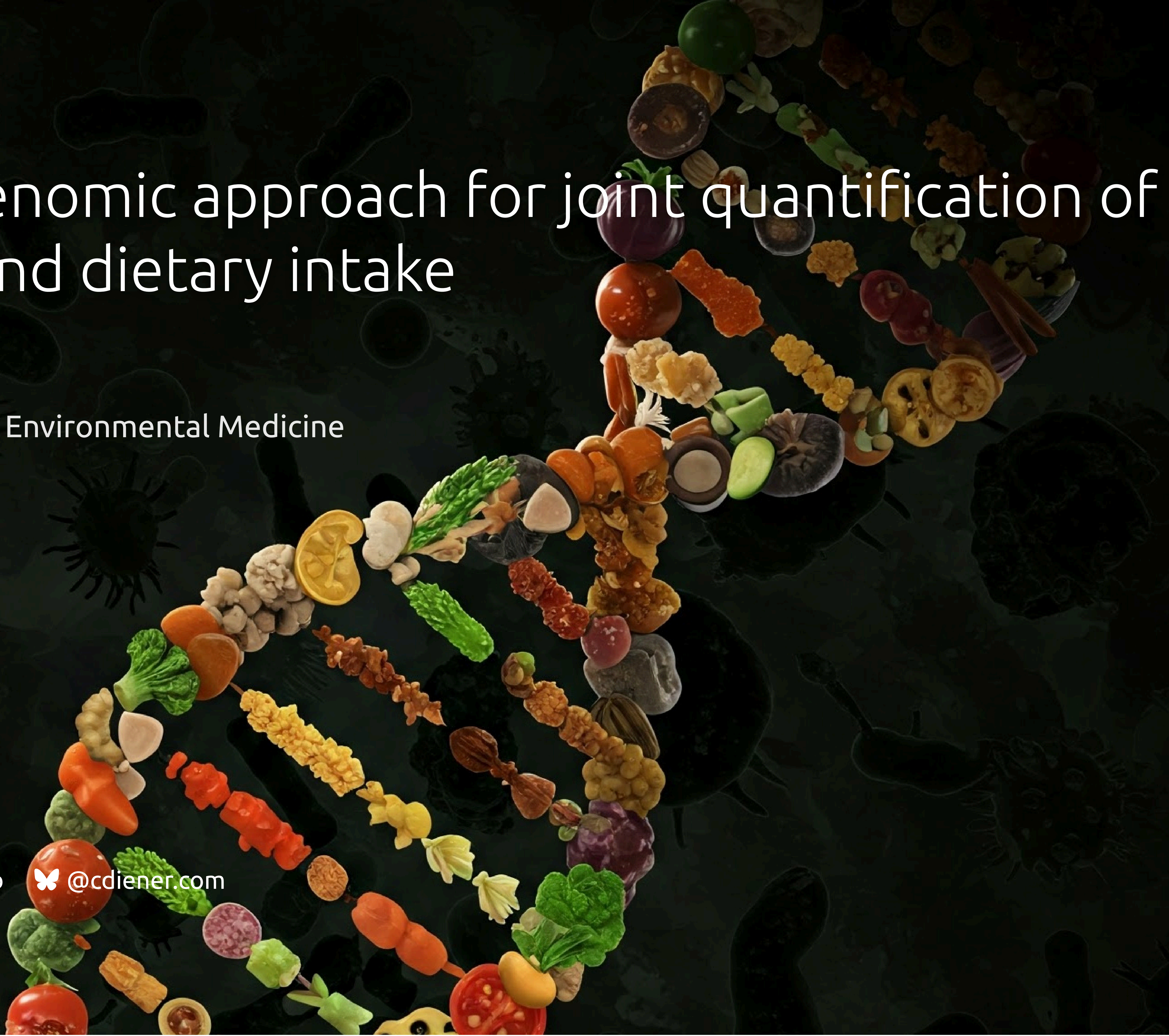

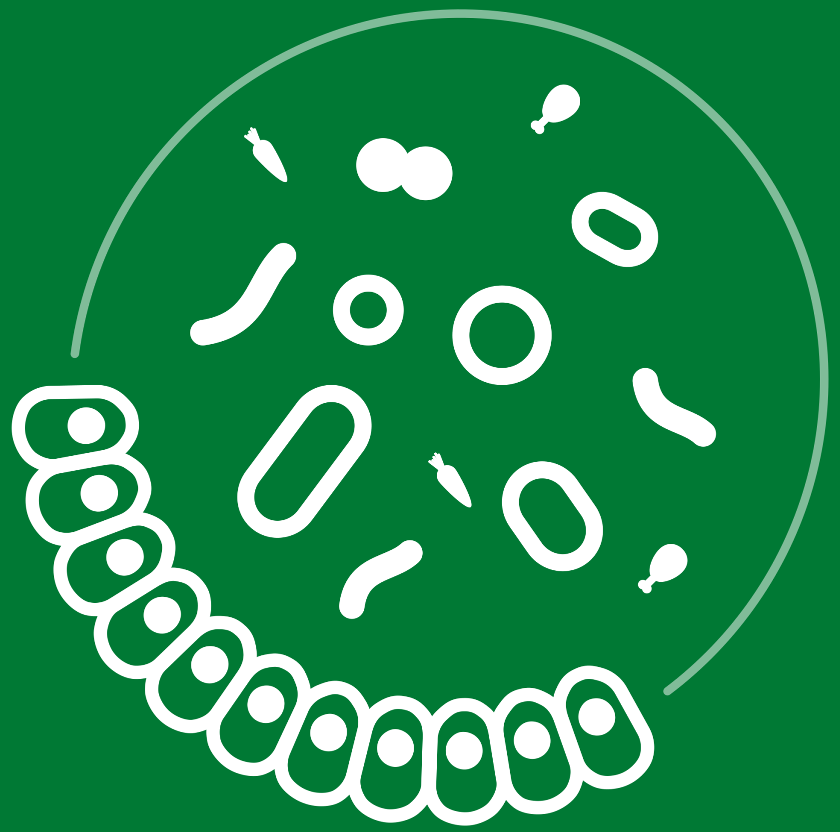

Metagenomic (total DNA) sequencing of fecal samples is widely used in microbiome studies.

Many foods we consume contain highly specific DNA sequences as well.

DNA can potentially comply with the requirements of a biomarker and there are some established marker genes (*trnL* or 12S rRNA gene).

Can we leverage total fecal DNA to measure food intake?

C Cuparencu et al. 2024, Nature Metabolism, https://doi.org/10.1038/s42255-024-01067-y

BL Petrone et al. 2023, PNAS, https://doi.org/10.1073/pnas.2304441120

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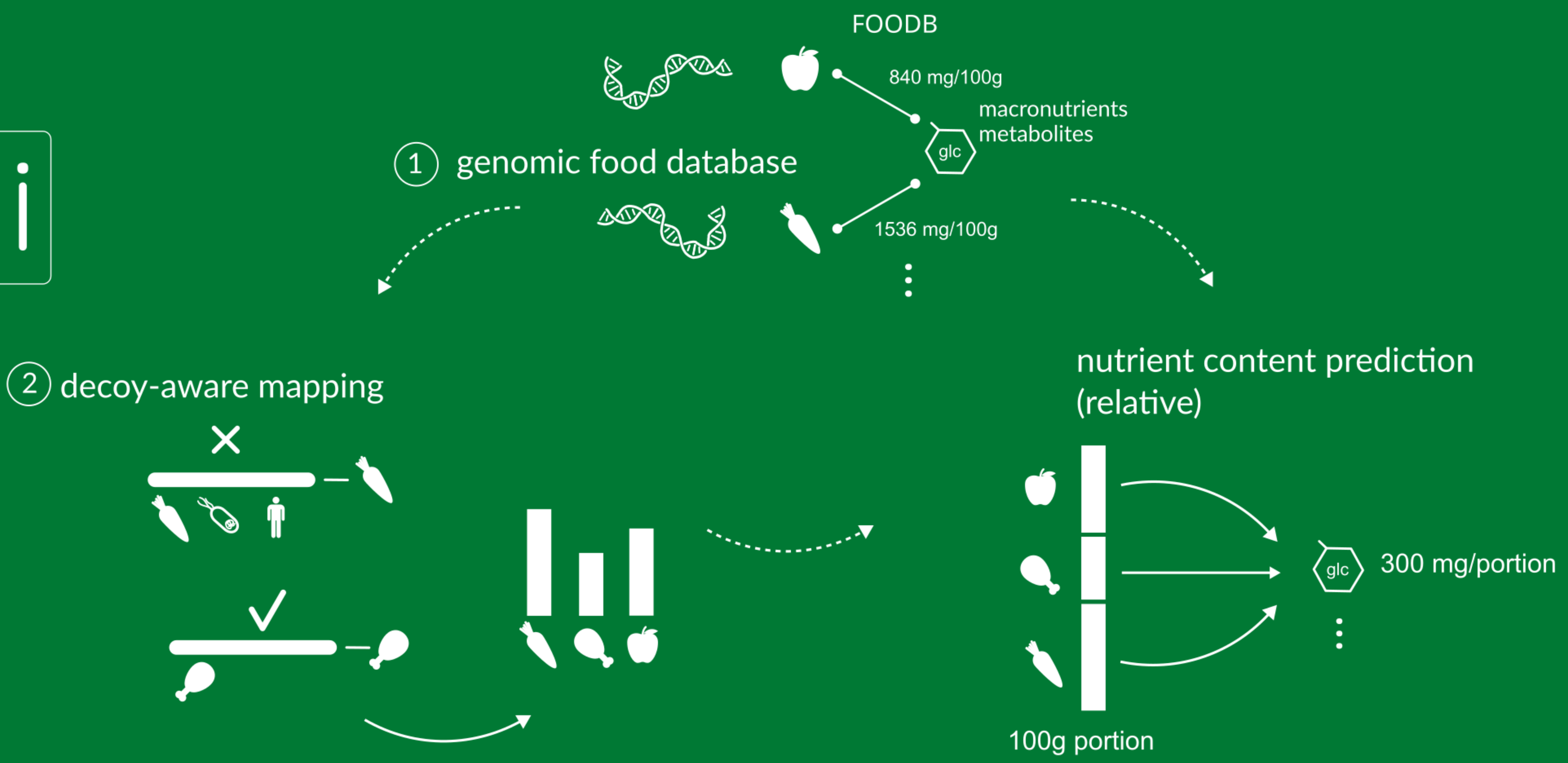
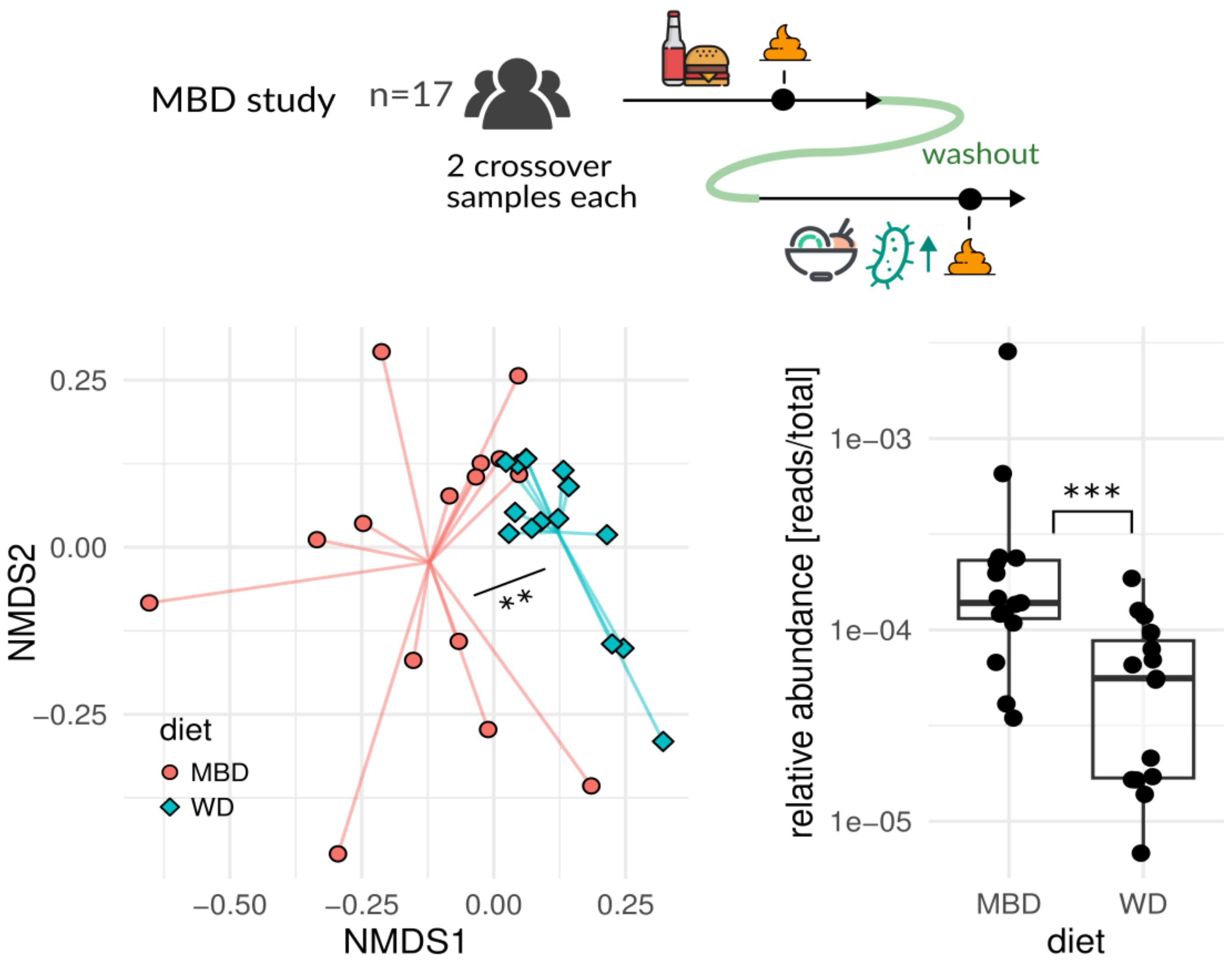


Challenges lie in the lack of **food-specific genome databases** and the **low biomass** of foods in fecal samples compared to microbes. MEDI attempts to solve those.

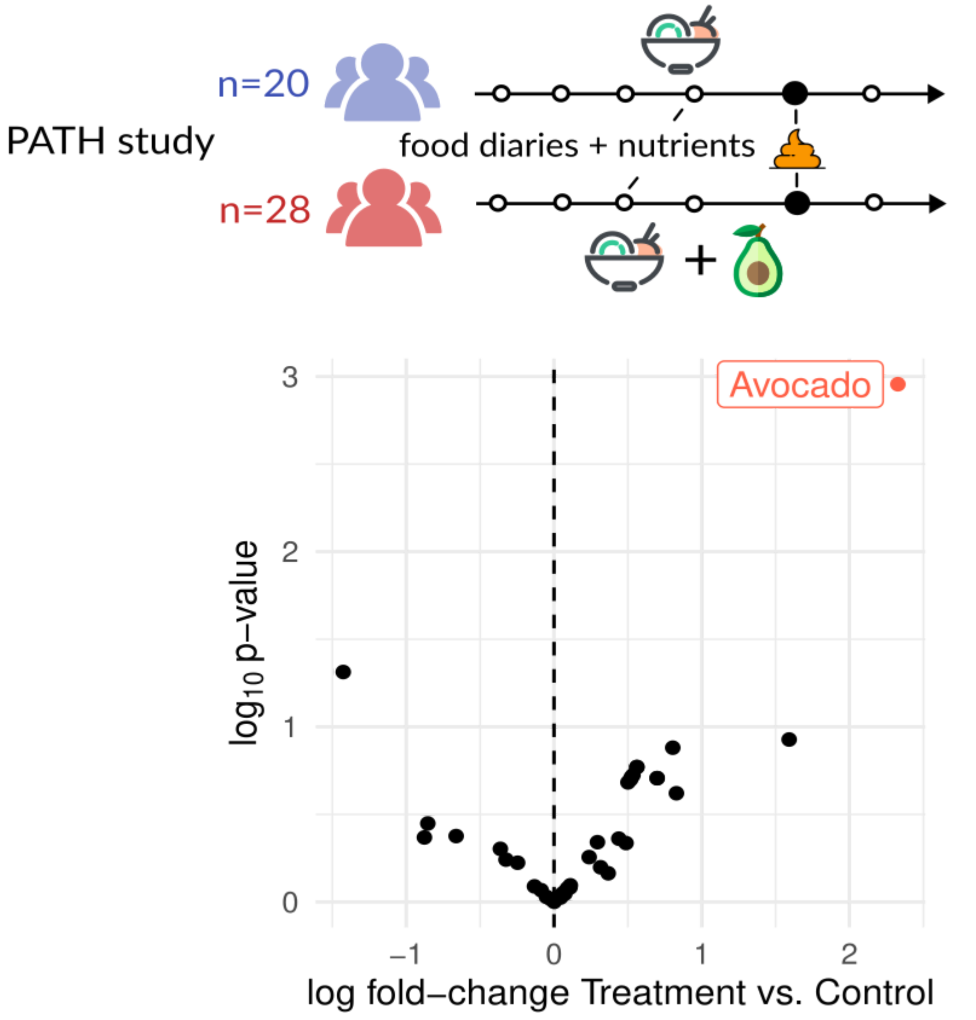
A database of food DNA database linked to nutrients

A decoy-aware mapping strategy

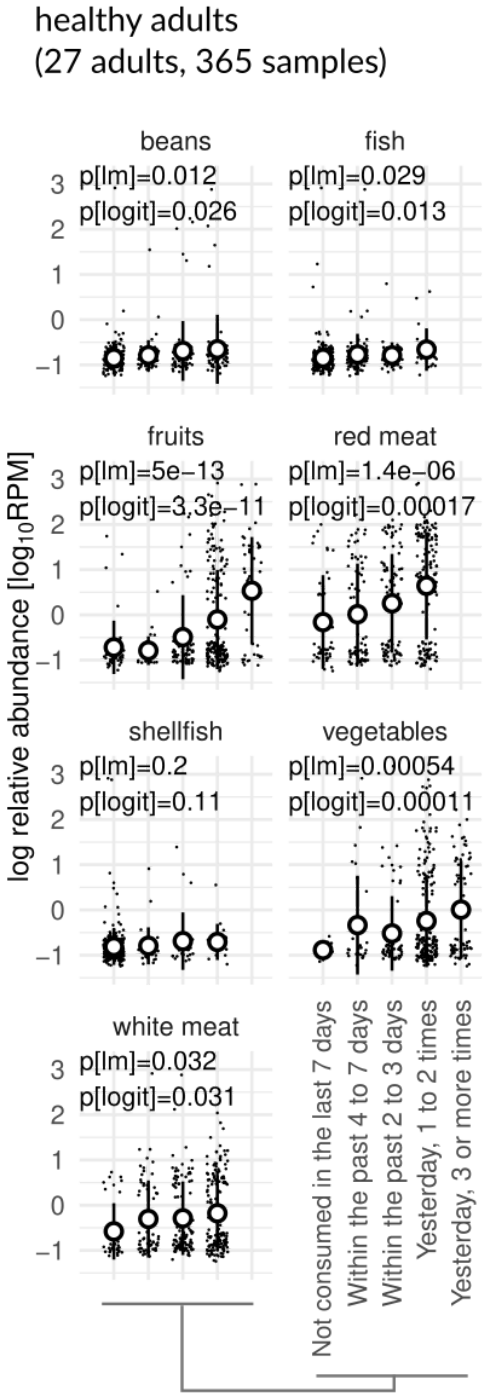
Validation with controlled-feeding studies and FFQs



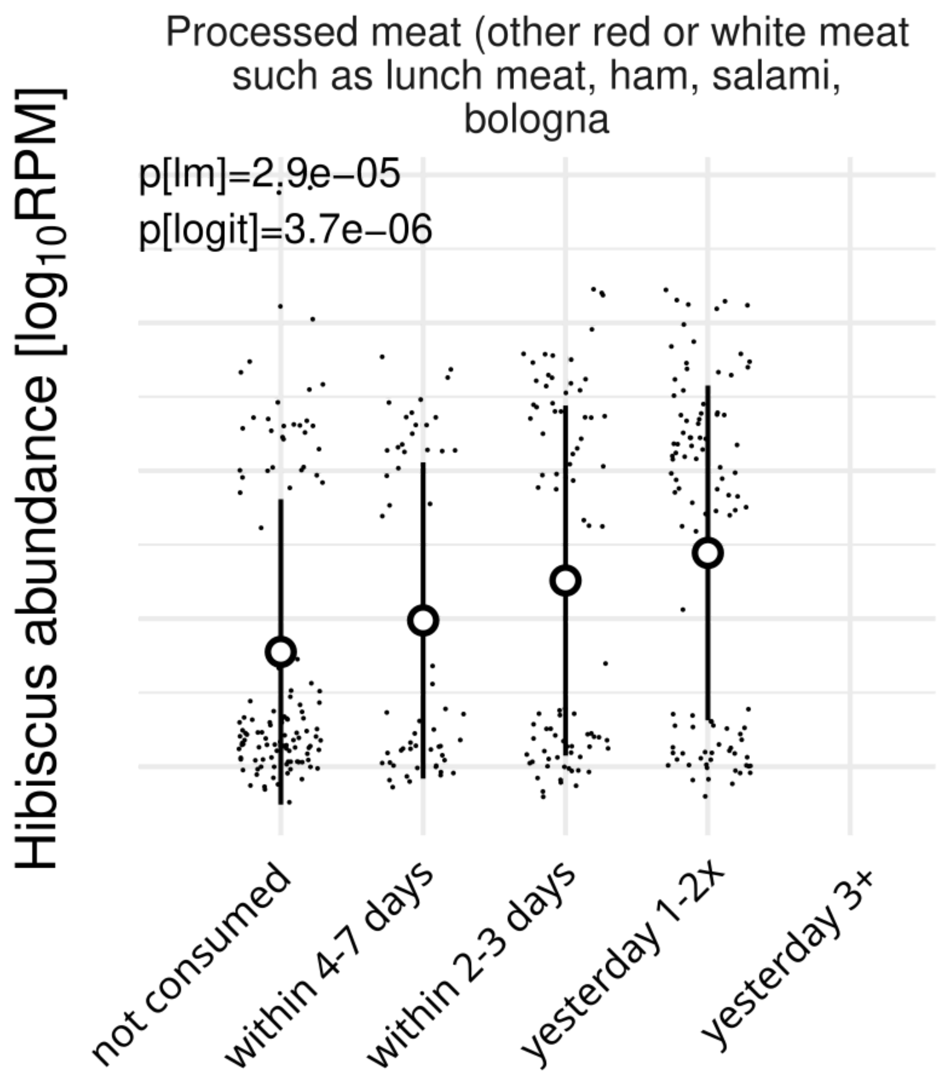
data provided by Karen Corbin

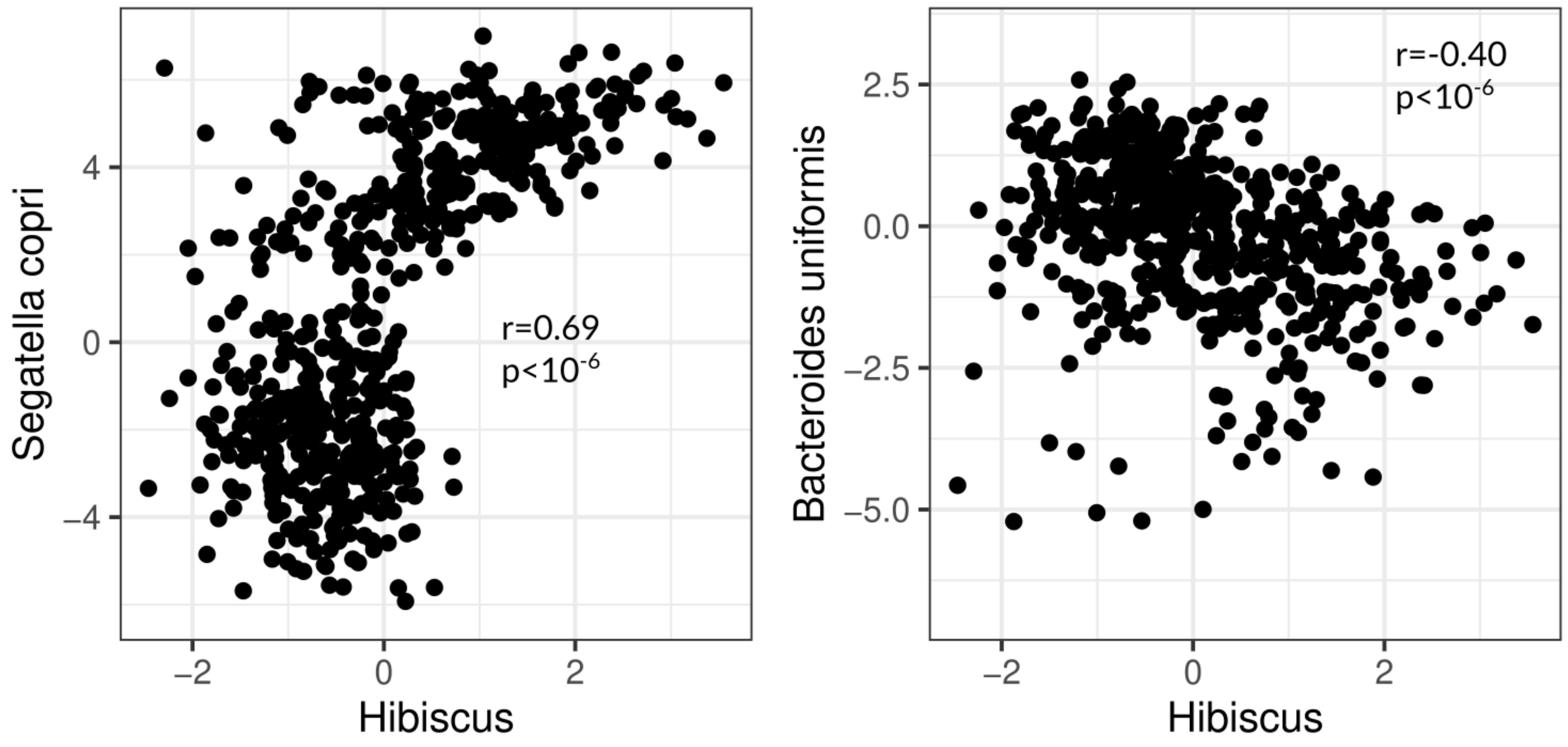


data provided by Hannah Holscher



A processed food signal associated with the gut microbiome





SECOM r and p-values, bias-corrected abundances

May be related to carboxymethylcellulose (CMC) levels
https://doi.org/10.1101/2025.02.21.639485

Thanks! 😄

MedUni Graz
Klara Filek
Christine Moissl-Eichinger

U of I Urbana-Champaign
Hannah Holscher

AdventHealth
Karen Corbin

ISB
Sean Gibbons

Funding

FWF Austrian Science Fund
Cluster of Excellence
Microbiomes Drive Planetary Health
Join our growing community!

NIH National Institute of Diabetes and Digestive and Kidney Diseases

C Diener et al. 2025, Nature Metabolism, https://doi.org/10.1038/s42255-025-01220-1

Take home messages

Strengths:

- high specificity (down to strain level)
- quantitative for foods and semi-quantitative for nutrients
- nice for retrospective studies
- high throughput

Limitations:

- cannot distinguish preparation types or site
  → resolve with proteomics or metabolomics?
- bias towards non-digested food
- sensitivity limited by low food DNA proportion
  → deeper sequencing?
- detection dependent on consumption time
  → pooling or time series?